

1) what is React?

Ans:- * React is a JavaScript lib used to build fast user interfaces especially for web applications.

* It helps create fast int websites using Components.

2) IS React a library or a framework?

Ans:- * React is a library, not framework.

3) key features of Reacts?

Ans:-

* Components - based arch

* Virtual DOM (fast update)

* Reusable code.

* one-way data flow

* JSX Syntax.

4) what is a Single Page Application (SPA)?

Ans:- SPA means only one HTML page loads once, then content updates dynamically.

Ex:- * No full page reload.
* fast user experience.

5) what is components in React?

Ans:- * A components is a small reusable piece of UI.

Ex:- * Header * footer * Button.

6) what is functional components?

Ans:- * Functional components

Return UI.

```
function Hello () {  
  return <h1> Hello </h1>;  
}
```


* mostly used in modern React

7) what is JSX?

Ans:- * JSX $\Rightarrow$ JavaScript XML

* Looks like HTML inside JavaScript.

* used to write UI easily

Ex:- `const element = <h1>Hello</h1>`

8) Can browsers understand JSX directly?

Ans:- * NO

* Browsers understand JavaScript not JSX.

9) What is Babel in React?

Ans:- * Babel is a Compiler

converts JSX to JavaScript

* So browser can understand the code

o) what is virtual DOM?

Ans:- * Virtual DOM is lightweight copy of real DOM.

* React updates Virtual DOM first.

* Then updates only changed parts in real DOM

* faster Performance.

Component & Props Questions.

1) what is the difference between functional and class components?

Ans:- 1. functional components:-

* simple javascript functions.

```
function Hello() {  
  return <h1> Hello2/h1>;  
}
```


* Easy to write

* uses Hooks (use state, use effect)

* Modern & recommended

2. Class Components :-

* Es6 classes.

```
class Hello extends React.Component
```

```
render () {
```

```
  return <h1>Hello</h1>;
```

```
}
```

```
}
```

* uses this keyword.

* Has lifecycle methods.

* Older approach.

Key Difference:-

Feature	Functional Component	Class Component
Syntax	Functions	Class
State	useState Hook	this.state.
Life cycle	useEffect Hook	lifecycle method
Code	Simple	Complex
* usage	Modern React	Old React.

Imp:-

* nowadays, developers mostly use functional components instead of class components.

Interview :-

functional components are simple function using Hooks, while class components are ES6 classes using state and lifecycle methods.